Number crunch at the print table

Fingerprints

Name _____ Date _____

Regroup tens and ones while you count paper ridge marks, cards, and magnifiers.

1. Little draws 47 curved ridge marks, then adds 26 more. How many ridge marks are in the drawing?

47 + 26 = _____

Show your thinking.

2. There are 68 paper pattern cards. Big puts away 29. How many cards remain on the table?

68 − 29 = _____

Show your thinking.

3. A tray holds 36 loop sketches and 58 whorl sketches. How many sketches are there altogether?

36 + 58 = _____

Show your thinking.

Number crunch at the print table

Fingerprints

Name _____ Date _____

Regroup tens and ones while you count paper ridge marks, cards, and magnifiers.

4. Little circles 46 details on a sheet with 72 ridge details. How many details are not circled?

72 − 46 = _____

Show your thinking.

5. Big sets out 59 cream print cards and 24 tan print cards. How many cards are ready?

59 + 24 = _____

Show your thinking.

6. The field table begins with 81 magnifier cards and uses 37. How many magnifier cards are left?

81 − 37 = _____

Show your thinking.

WONDER DAILY

Fingerprints

A mark on the glass

Morning light revealed a faint swirl on a glass. Under the magnifier, Little saw many curved lines instead of one plain smudge.

Ridges and grooves

A fingertip has raised friction ridges and lower grooves. The ridge tops touched the glass and left separate curved lines.

Three broad flows

Loops curve back, whorls circle a center, and arches rise across. These families describe the broad flow, not every tiny detail.

Inside the loop

Two prints were both loops, yet their small details differed. Ridges ended and split in different places across each map.

A print on paper

Washable ink coated the raised ridge tops. They made dark paths on paper while the lower grooves left pale spaces.

Same map, new mark

A lighter press looked smudged, but its turns, endings, and splits still lined up. Pressure changed the mark, not the ridge map.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Describe a ridge map

Fingerprints · Curiosity day

Name _____ Date _____

Choose a close look and compare the paths you see.



Three broad flows



Tiny turns and forks



A careful second look

Draw two large ridge paths. Add one ending and one split, then circle what changed.

ridge · groove · loop · whorl · arch · ending · split · impression · detail · evidence

Both patterns … name one feature they share

One ridge … tell where it ends or splits

The evidence shows … connect a detail to your conclusion

Did you name a detail and explain what that detail helped you notice?

Fingerprints

01 What made the curved lines on the glass?



A fingertip can leave a map of its raised ridges.

Look closer. Look for the pale ridges and the darker spaces between them.

A little explanation. A fingertip has raised lines called friction ridges, with lower grooves between them. When ridge tops touch a surface, sweat and natural skin oils can leave a faint impression of their paths.

Try this. Tilt a clean drinking glass toward a lamp and look for a faint print. Keep the glass on the table while you look.

02 Which way do the ridges flow?



Loops, whorls, and arches describe the broad ridge flow.

Look closer. Trace one curve in each large pattern card with your eyes.

A little explanation. Many fingerprints can be grouped by their broad ridge flow. A loop curves back toward one side, a whorl circles around a center, and an arch rises and crosses from one side to the other.

Try this. Draw one loop, one circle-like whorl, and one hill-shaped arch on paper.

03 Can two loops still be different?



The same broad pattern can hold different tiny details.

Look closer. Find a ridge that ends and another that splits into two paths.

A little explanation. The big pattern family is only a first clue. Along the ridges are smaller details: a line may end, split, shorten, or bend. Two prints can both be loops while their tiny details appear in different places.

Try this. On two drawn loops, add endings and forks in different places. Compare your maps.

04 Why does a print have dark lines and pale gaps?



Ridge tops transfer ink; lower grooves leave pale gaps.

Look closer. Compare the inked fingertip with the print it leaves on paper.

A little explanation. Washable ink touches the raised ridge tops. When the fingertip presses paper, those tops transfer dark lines while the lower grooves leave pale spaces. The print is a flat impression of a textured fingertip.

Try this. Use a non-toxic washable stamp pad on one fingertip, press once on paper, and wash your hands. Keep the paper at home.